



Pizza Hut Sales Analysis using SQL

PIZZA

ABOUT

About the Project

Pizza Hut Sales Analysis using SQL is a data analytics project designed to analyze sales transactions and generate meaningful business insights using Structured Query Language (SQL). The project involves working with a relational database containing customer orders, order details, pizza information, and pizza categories to understand sales performance and customer purchasing behavior.

The analysis focuses on identifying key business metrics such as total revenue, order distribution, best-selling pizzas, category-wise sales, revenue contribution, and peak ordering hours. To perform these analyses, various SQL techniques including JOINS, Aggregate Functions, GROUP BY, ORDER BY, Subqueries, Common Table Expressions (CTEs), and Window Functions are applied.

This project demonstrates the practical application of SQL in solving real-world business problems by transforming raw transactional data into actionable insights. It highlights the importance of data analysis in supporting business decisions, optimizing product performance, and understanding customer demand.



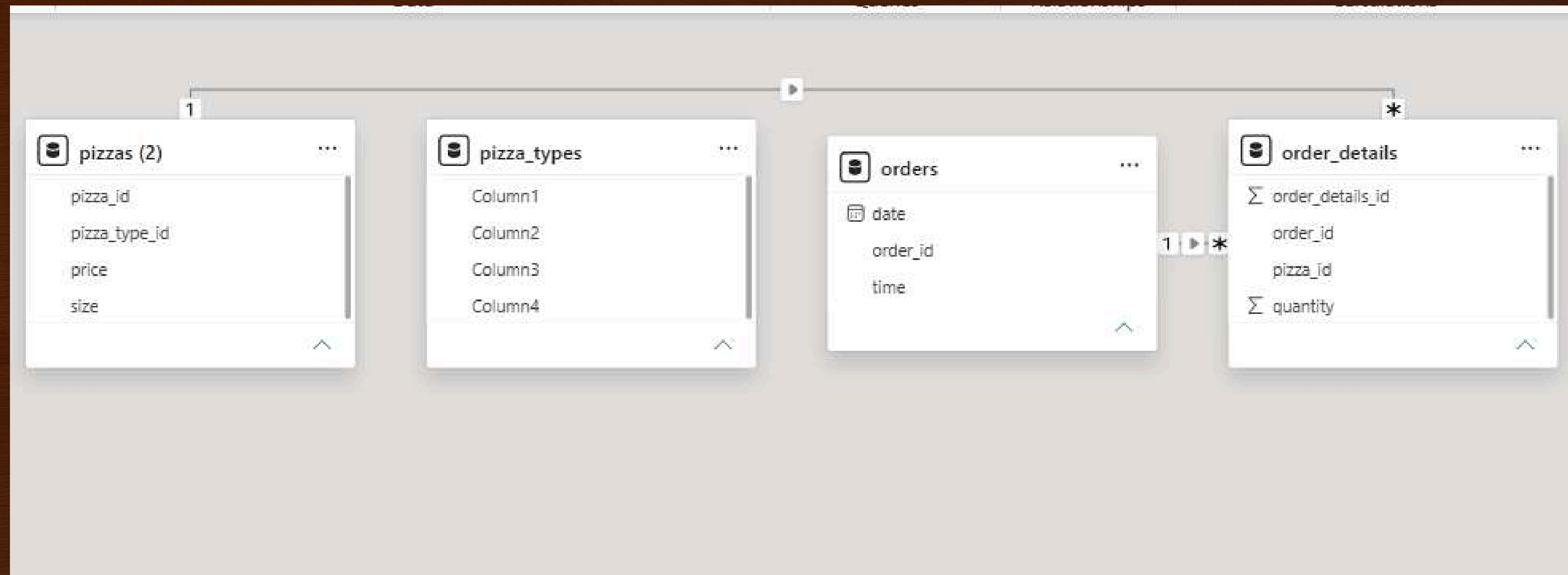
DATABASE MODEL

1

The Pizza Hut database consists of four interconnected tables: orders, order_details, pizzas, and pizza_types. These tables are related through primary and foreign keys, ensuring efficient data storage and accurate sales analysis.

2

The data model defines the relationships between customer orders, pizza details, and product categories. It enables SQL queries to analyze revenue, sales trends, order quantity, and overall business performance.

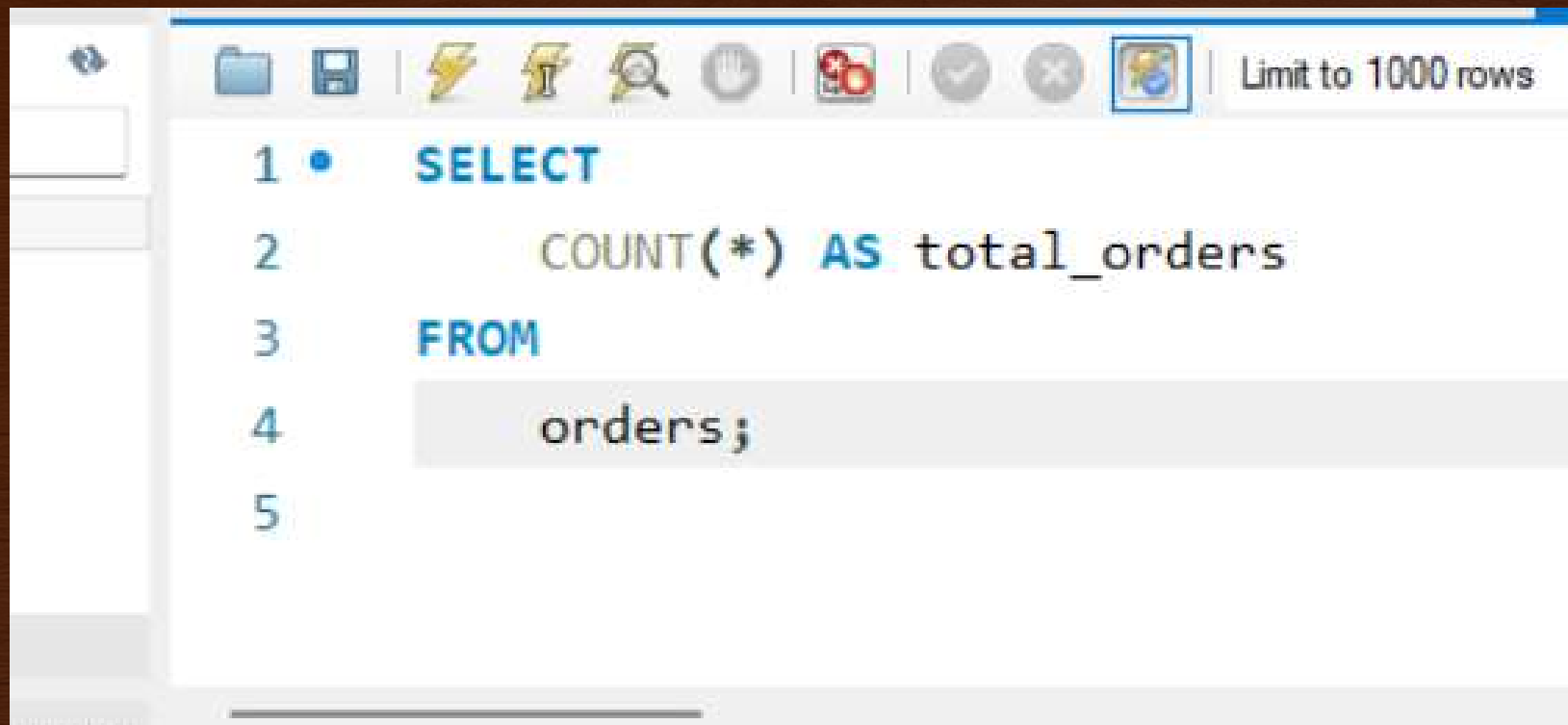


SQL QUERY 01

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED

Objective:

To determine the total number of orders placed by customers during the recorded sales period.



A screenshot of a SQL query editor window. The toolbar at the top includes icons for file operations, execution, and a 'Limit to 1000 rows' option. The query text is as follows:

```
1 • SELECT
2     COUNT(*) AS total_orders
3 FROM
4     orders;
5
```



A screenshot of the 'Result Grid' window. It displays a single row of results for the query. The column header is 'total_orders' and the value is '21350'.

	total_orders
▶	21350

SQL QUERY 02

CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES

Objective:

To calculate the total revenue generated from all pizza sales by multiplying the quantity sold with the price of each pizza.

```
SELECT  
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS total_sales  
FROM  
    order_details  
    JOIN  
    pizzas ON pizzas.pizza_id = order_details.pizza_id
```

Result Grid	
	total_sales
▶	817860.05

SQL QUERY 03

IDENTIFY THE HIGHEST-PRICED PIZZA.

Objective:

To identify the pizza with the highest selling price available in the Pizza Hut menu.

```
• SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	

SQL QUERY 04

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

Objective:

To determine the pizza size that is ordered most frequently by customers.

This analysis helps understand customer preferences and supports inventory and production planning.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

	size	order_count
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

SQL QUERY 05

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

Objective

To identify the top 5 most ordered pizza types based on the total quantity sold:

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

	name	quantity
►	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

SQL QUERY 06

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

Objective

Join the necessary tables to find the total quantity of each pizza category ordered.

```
• SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
|
```

	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

SQL QUERY 07

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

Objective

To analyze the distribution of customer orders across different hours of the day.

```
SELECT
    HOUR(order_time) AS hours, COUNT(order_id) AS total_orders
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid				 Filter Rows:
	hours	total_orders		
▶	11	1231		
	12	2520		
	13	2455		
	14	1472		
	15	1468		
	16	1920		
	17	2336		
	18	2399		
	19	2009		
	20	1642		

SQL QUERY 08

JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS

Objective

To analyze the distribution of pizzas across different categories.

```
select category ,count(name) from pizza_types  
group by category
```

Result Grid			Filter Rows
	category	count(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

SQL QUERY 09

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

Objective

To calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0) AS order_per_day
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

	order_per_day
▶	138

SQL QUERY 10

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

Objective

To determine the top 3 pizza types based on total revenue generated.

```
select pizza_types.name,  
sum(order_details.quantity * pizzas.price) as revenue  
from pizza_types join pizzas  
on pizzas.pizza_type_id = pizza_types.pizza_type_id  
join order_details  
on order_details.pizza_id = pizzas.pizza_id  
group by pizza_types.name order by revenue desc limit 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

SQL QUERY 11

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

Objective

To calculate the percentage contribution of each pizza category to the total revenue.

```
SELECT
    pizza_types.category,
    round((SUM(order_details.quantity * pizzas.price) / (SELECT
        ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
    FROM
        order_details
        JOIN
            pizzas ON pizzas.pizza_id = order_details.pizza_id)) * 100 ,2 )AS revenue
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

	category	revenue
►	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68

SQL QUERY 12

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

Objective

To analyze the cumulative revenue generated over time.

```
select order_date,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(order_details.quantity * pizzas.price)as revenue  
from order_details join pizzas  
on order_details.pizza_id = pizzas.pizza_id  
join orders  
on orders.order_id = order_details.order_id  
group by orders.order_date)as sales;
```

	order_date	cum_revenue
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.3500000000002
	2015-01-11	25862.65
	2015-01-12	27781.7

SQL QUERY 13

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

Objective

To identify the top 3 revenue-generating pizza types within each pizza category.

```
(select category,name,revenue,
rank() over(partition by category order by revenue desc)as rn
from
(select pizza_types.category,pizza_types.name,
sum((order_details.quantity) * pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category , pizza_types.name) as a) as b
where rn <= 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5
	The Classic Deluxe Pizza	38180.5
	The Hawaiian Pizza	32273.25
	The Pepperoni Pizza	30161.75
	The Spicy Italian Pizza	34831.25
	The Italian Supreme Pizza	33476.75
	The Sicilian Pizza	30940.5
	The Four Cheese Pizza	32265.700000000065
	The Mexicana Pizza	26780.75

SKILL DEMONSTRATED

- SQL Query Writing
- Data Analysis
- Relational Database Management
- Data Modeling
- Business Insight Generation
- Problem Solving
- MySQL
- Power BI (Data Model)



CONCLUSION

This project demonstrates the practical application of SQL in analyzing real-world sales data. By performing data extraction, aggregation, joins, and revenue analysis, meaningful business insights were generated to support data-driven decision-making. The project enhanced my understanding of relational databases, SQL querying, and business analytics, while strengthening my ability to solve real-world data analysis problems.



FUTURE SCOPE

- Develop an interactive Power BI dashboard.
- Perform predictive sales analysis using Python.
- Build automated sales reports.
- Analyze customer purchasing behavior using machine learning.

ABOUT THE AUTHOR

I am Rituraj Singh Chouhan, a B.Tech Computer Science and Engineering student at IPS Academy, Indore. I am passionate about Data Analytics, SQL, Power BI, Python, and Generative AI. I enjoy analyzing real-world data and building projects that solve business problems through data-driven insights. I am continuously learning and improving my technical skills to build a successful career as a Data Analyst.

- Email: riturajsinghofficial237@gmail.com
- LinkedIn: [linkedin.com/in/rituraj-singh-171b8a2b7](https://www.linkedin.com/in/rituraj-singh-171b8a2b7)
- GitHub: github.com/Riturajsingh07

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THANK YOU

This project demonstrates my understanding of SQL, Relational Databases, Data Analysis, and Business Intelligence through the analysis of the Pizza Hut Sales Dataset. I hope this report provides meaningful insights into my technical skills and problem-solving approach.